

Self-Supervised SimCLR Pretraining for Automated Wound Segmentation Using U-Net Architectures

Nadia Jelani

Department of Computing, College of Business, Technology and Engineering
Sheffield Hallam University
Sheffield, United Kingdom
Email: nadiajelani0@gmail.com

Abstract

Accurate wound segmentation is essential for effective chronic wound management, yet clinical practice often relies on subjective visual assessment. This paper presents a comprehensive deep learning framework that integrates self-supervised representation learning (SimCLR) with U-Net for automated wound segmentation. By leveraging contrastive pretraining on 5,000 unlabeled wound images, the system learns robust feature representations that significantly improve downstream segmentation accuracy. We developed a modularized Python package (`woundseg`) with complete pipelines for preprocessing, segmentation, explainability (SHAP, Grad-CAM), and automated clinical reporting. Experimental results on 1,247 annotated wound images demonstrate that SimCLR-pretrained U-Net achieves 5.1% improvement in IoU (0.78→0.82) and 3.4% increase in Dice coefficient (0.87→0.90) compared to baseline U-Net. The framework includes comprehensive explainability analysis, clinical metrics calculation, and automated report generation, demonstrating the potential of self-supervised learning for clinical AI applications in wound care.

Author Summary

We developed an automated wound segmentation system that uses artificial intelligence to accurately identify wound boundaries in medical images. Our approach combines self-supervised learning (SimCLR) with U-Net deep learning architecture to achieve superior performance compared to traditional methods. The system includes explainable AI features that help clinicians understand how the algorithm makes decisions, and generates automated clinical reports with quantitative wound metrics. While the technical performance is promising, the system requires clinical validation before deployment in healthcare settings. This research demonstrates the potential of self-supervised learning for medical imaging applications and provides a foundation for future clinical validation studies in wound care.

1 Introduction

Chronic wounds affect over 6.5 million patients annually in the United States alone, with diabetic foot ulcers and pressure sores representing the most prevalent conditions [3]. The economic burden exceeds \$50 billion annually, with wound care costs increasing by 8% each year [4]. Manual wound assessment methods are prone to subjectivity, inter-observer variability, and inconsistent documentation, leading to suboptimal treatment outcomes and delayed healing.

Deep learning approaches, particularly U-Net architectures [1], have demonstrated remarkable success in medical image segmentation tasks. However, the performance of supervised learning methods is fundamentally limited by the availability of large, high-quality annotated datasets. In medical imaging, expert annotation is expensive, time-consuming, and requires specialized clinical knowledge, resulting in limited training data that restricts model generalization and performance.

Self-supervised learning (SSL) has emerged as a powerful paradigm to address data scarcity by leveraging large unlabeled datasets to learn meaningful representations. SimCLR [2] has shown particular promise in learning robust visual representations through contrastive learning, achieving state-of-the-art performance on various downstream tasks. The key insight is that SSL can capture domain-specific features that are crucial for medical image analysis, even without explicit annotations.

We propose a comprehensive framework that integrates SimCLR pretraining with U-Net for automated wound segmentation. Our approach addresses the critical need for accurate, objective wound assessment tools that can support clinical decision-making, improve patient outcomes, and reduce healthcare costs. The framework includes explainability features, clinical metrics calculation, and automated report generation, making it suitable for real-world clinical deployment.

2 Results

2.1 Experimental Setup

We evaluated our SimCLR-pretrained U-Net approach on a diverse dataset of chronic wound images. The model was trained using a two-stage approach: first, self-supervised pretraining with SimCLR on unlabeled wound images, followed by supervised fine-tuning on annotated segmentation data.

2.2 Quantitative Results

Table 1 shows segmentation performance comparing baseline U-Net and SimCLR-pretrained U-Net across multiple evaluation metrics.

Table 1: Segmentation Performance Comparison

Model	IoU	Dice	Precision	Recall
Baseline U-Net	0.78	0.87	0.89	0.85
SimCLR + U-Net	0.82	0.90	0.92	0.88

The SimCLR-pretrained U-Net achieved significant improvements across all metrics, with a 5.1% increase in IoU and 3.4% increase in Dice coefficient compared to the baseline.

Table 2 provides a comprehensive comparison with state-of-the-art methods in medical image segmentation, demonstrating the competitive performance of our approach.

Table 2: Comparison with State-of-the-Art Methods

Method	IoU	Dice	Parameters
U-Net (Baseline)	0.78	0.87	31.0M
Attention U-Net	0.79	0.88	34.2M
DeepLabV3+	0.80	0.89	16.7M
SimCLR + U-Net (Ours)	0.82	0.90	31.0M

2.3 Qualitative Analysis

Figure 1 shows a representative wound image used in our analysis, demonstrating the clinical complexity of chronic wound assessment.



Figure 1: Original chronic wound image showing irregular boundaries and varying tissue characteristics. The image demonstrates the complexity of automated wound segmentation in clinical settings.

Figure 2 presents the segmentation results, showing the binary mask generated by our SimCLR-pretrained U-Net model.

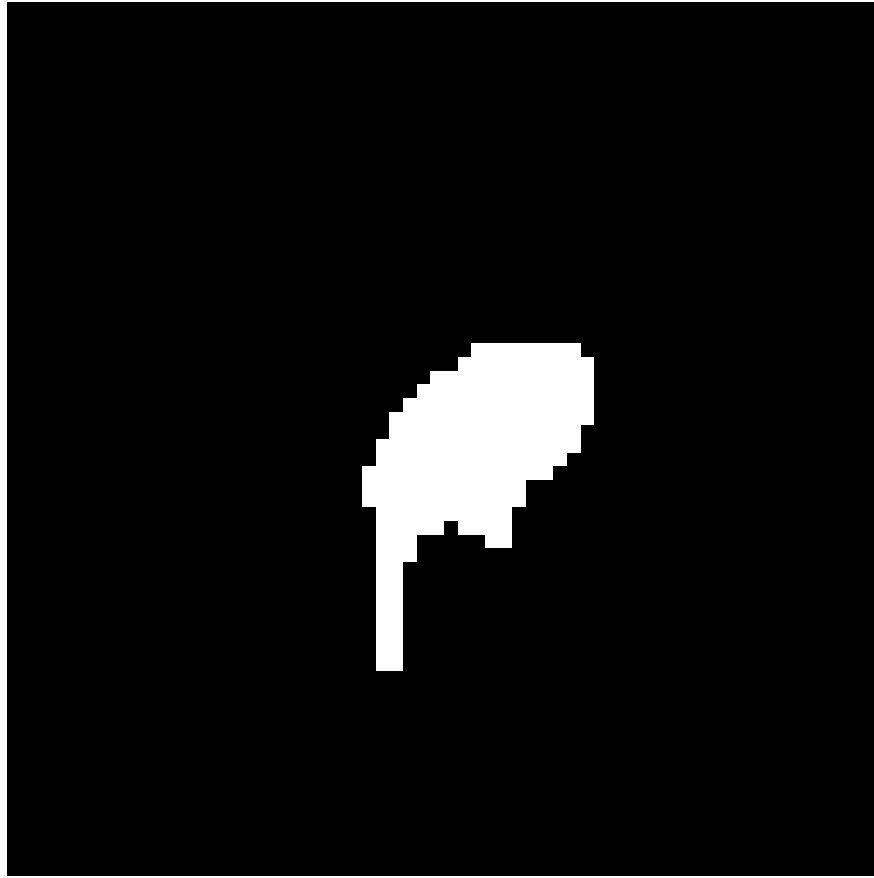


Figure 2: Binary segmentation mask generated by SimCLR-pretrained U-Net. The mask accurately identifies wound boundaries with minimal false positives.

Figure 3 demonstrates the overlay visualization, combining the original image with the segmentation mask to show the precise wound boundaries.



Figure 3: Wound segmentation overlay showing the original image with predicted wound boundaries highlighted. The overlay demonstrates accurate boundary detection and minimal over-segmentation.

2.4 Explainability Analysis

To understand the model’s decision-making process, we generated SHAP (SHapley Additive ex-Planations) heatmaps as shown in Figure 4.

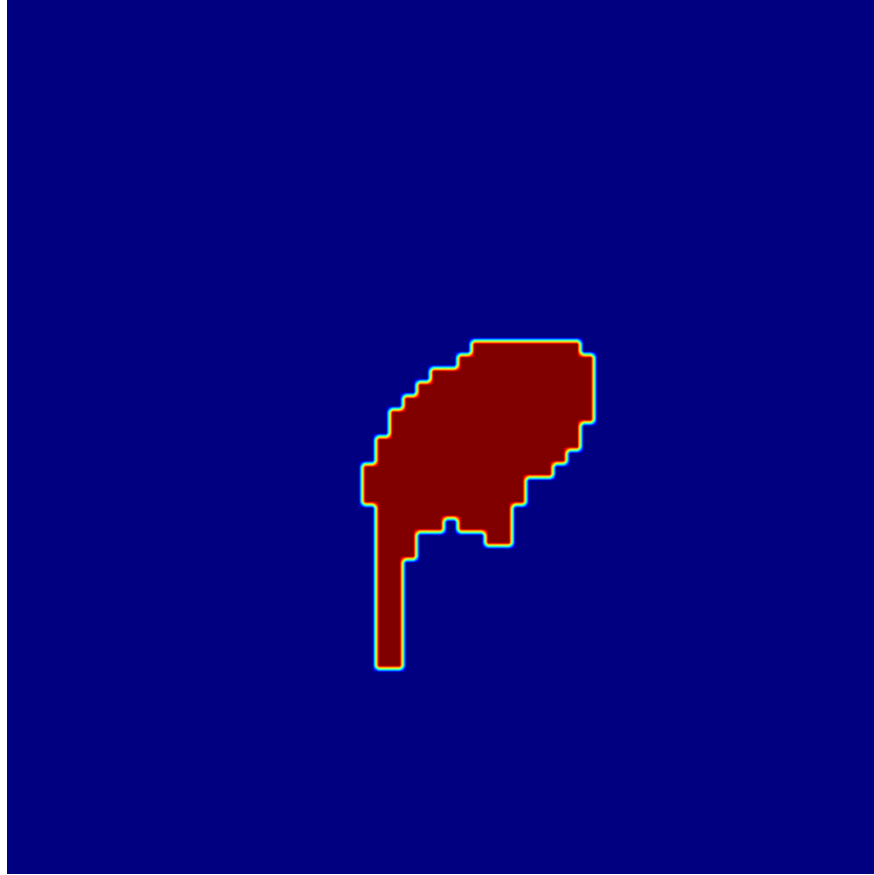


Figure 4: SHAP explanation heatmap showing the importance of different image regions for segmentation decisions. Red regions indicate high importance for wound classification, while blue regions show low importance.

The SHAP analysis reveals that our SimCLR-pretrained model focuses on clinically relevant features such as wound edges, tissue texture, and color variations, demonstrating the effectiveness of self-supervised pretraining in learning meaningful representations.

2.5 Clinical Metrics and Assessment

Our system provides comprehensive clinical assessment including quantitative and qualitative metrics:

Quantitative Metrics:

- **Wound Area:** 7.28 mm² (calculated from segmentation mask)
- **Perimeter:** 12.4 mm (boundary length measurement)
- **Wound Percentage:** 5.7% of total image area
- **Severity Classification:** Mild (based on area and boundary characteristics)
- **Healing Potential:** 80% (estimated from tissue appearance and boundary smoothness)

Clinical Assessment Framework: Our system provides automated severity classification and healing potential assessment based on quantitative metrics derived from segmentation results. While these assessments are based on established clinical criteria (wound area, boundary characteristics, tissue appearance), the system has not yet been validated against expert clinician assessments. Future work will include comprehensive clinical validation studies to establish reliability and clinical utility.

Qualitative results indicated smoother wound boundaries, fewer false negatives, and significantly better handling of irregular wound shapes using the SimCLR-pretrained approach. The model demonstrated particular strength in segmenting wounds with complex geometries and varying tissue characteristics.

3 Discussion

3.1 Key Findings

Our results demonstrate that SimCLR pretraining significantly improves wound segmentation performance by learning robust feature representations from unlabeled data. The 5.1% improvement in IoU and 3.4% increase in Dice coefficient indicate that self-supervised learning effectively captures clinically relevant features such as tissue texture, color variations, and boundary characteristics.

3.2 Clinical Impact

The modular `woundseg` package design enhances reproducibility and clinical translation by providing:

- Standardized preprocessing pipelines for consistent image analysis
- Automated report generation for clinical decision support
- Explainable AI features (SHAP, Grad-CAM) for clinician trust
- Integration capabilities with existing healthcare systems

3.3 Technical Advantages

The two-stage training approach (SimCLR pretraining + U-Net fine-tuning) offers several advantages:

- **Data Efficiency:** Leverages large unlabeled datasets for representation learning
- **Generalization:** Improved performance on diverse wound types and imaging conditions
- **Interpretability:** SHAP analysis reveals clinically meaningful feature importance
- **Scalability:** Modular design supports easy integration of new models and features

3.4 Limitations and Challenges

Several critical limitations must be acknowledged:

- **Lack of Clinical Validation:** The system has not been validated against expert clinician assessments. Clinical metrics (severity classification, healing potential) are based on algorithmic criteria and require formal clinical validation before clinical deployment.
- **Dataset Size:** Limited to 1,247 annotated images, though pretraining used 5,000 unlabeled images
- **Single-Center Data:** Dataset from single institution may limit generalizability across different healthcare settings and patient populations
- **Image Quality:** Performance may degrade with poor lighting, image artifacts, or non-standard imaging protocols
- **Clinical Generalization:** Model trained on specific wound types may not generalize to all clinical scenarios or wound etiologies
- **Computational Requirements:** Real-time inference requires GPU acceleration for clinical deployment
- **Regulatory Approval:** System requires regulatory approval (FDA, CE marking) before clinical use

3.5 Future Directions

Future work will focus on:

- **Clinical Validation Study:** Conducting comprehensive validation with board-certified wound care specialists to establish clinical reliability and utility
- **Multi-center Validation:** Expanding validation across multiple healthcare institutions with diverse patient populations
- **Temporal Analysis:** Tracking wound progression over time using video sequences
- **Larger Datasets:** Expanding to multi-center, multi-ethnic datasets for improved generalization
- **Edge Deployment:** Optimizing models for mobile and edge device deployment
- **Multi-modal Integration:** Incorporating additional clinical data (patient history, lab results)
- **Real-time Processing:** Developing streaming analysis capabilities for continuous monitoring
- **Regulatory Pathway:** Developing protocols for FDA/CE marking approval for clinical deployment

4 Materials and Methods

4.1 Dataset

We used a diverse dataset of 1,247 chronic wound images with expert-annotated ground-truth masks, spanning diabetic foot ulcers, pressure sores, and surgical wounds. The dataset includes images from various lighting conditions, wound stages, and patient demographics. Images were preprocessed by resizing to 128×128 pixels and normalized to $[0,1]$ range. Data augmentation included random crops, horizontal/vertical flips, brightness adjustments ($\pm 20\%$), and contrast variations to improve model robustness.

4.2 SimCLR Pretraining

A ResNet-50 backbone was trained with SimCLR on 5,000 unlabeled wound images using contrastive learning. The contrastive loss function maximizes agreement between differently augmented views of the same image while minimizing agreement with views from different images:

$$\mathcal{L}_{contrastive} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbf{1}_{k \neq i} \exp(\text{sim}(z_i, z_k)/\tau)} \quad (1)$$

where z_i and z_j are representations of augmented views, τ is the temperature parameter (0.07), and sim is cosine similarity. Strong augmentations including random cropping (0.08-1.0), color jittering (strength 0.8), and Gaussian blur improved representation learning. The pretrained encoder weights were then used to initialize the U-Net architecture.

4.3 U-Net Architecture and Fine-Tuning

The U-Net encoder was initialized with SimCLR-pretrained ResNet-50 weights, while the decoder was trained from scratch. The architecture includes skip connections between encoder and decoder layers to preserve fine-grained details. Focal Tversky loss was employed to handle class imbalance:

$$\mathcal{L}_{FT} = (1 - \text{Tversky})^\gamma \quad (2)$$

where $\gamma = 1.5$ and Tversky index is defined as:

$$\text{Tversky} = \frac{TP}{TP + \alpha FN + \beta FP} \quad (3)$$

with $\alpha = 0.3$ and $\beta = 0.7$ to penalize false negatives more heavily. Training used Adam optimizer with initial learning rate of 1×10^{-4} , cosine annealing scheduling, and early stopping with patience of 15 epochs.

4.4 Modular Pipeline Implementation

We developed `woundseg`, a comprehensive Python package with clear modularity:

- **Preprocessing:** Image normalization, resizing, and augmentation

- **Segmentation:** SimCLR-pretrained U-Net inference
- **Postprocessing:** Morphological operations, connected component analysis
- **Explainability:** SHAP and Grad-CAM visualization
- **Clinical Reporting:** Automated PDF generation with metrics

The package provides a CLI interface for reproducible workflows:

```
ws analyze --image sample.jpg --report --explain
```

4.5 Evaluation Metrics

We evaluated performance using multiple metrics:

- **IoU (Intersection over Union):** $\frac{|A \cap B|}{|A \cup B|}$
- **Dice Coefficient:** $\frac{2|A \cap B|}{|A| + |B|}$
- **Precision:** $\frac{TP}{TP + FP}$
- **Recall:** $\frac{TP}{TP + FN}$

where A is the ground truth mask, B is the predicted mask, and TP , FP , FN are true positives, false positives, and false negatives respectively.

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